

1. The Elements of Machine Learning

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Contents

Title i

Copyright ii

Probabilistic Machine Learning 1

Appendix A: Probability Background 7

Probabilistic Machine Learning

A key concept in machine learning is that of *uncertainty*. It arises both through the noise in measurements, as well as through the finite size of data sets. Probability theory provides a consistent framework for the quantification and manipulation of uncertainty and forms one of the central foundations for machine learning. In what follows, we first review the basics of probability theory. We then show how it helps in capturing our uncertainty when learning models and making predictions within two schools of thought, i.e. frequentist and Bayesian statistics.

Probability Theory Concepts

A *random variable* is a variable whose value is subject to variation due to chance. For example, we can denote the outcome of tossing a coin by the random variable X whose values can be either H (for Heads) or T (for Tails); technically we say that the *domain* of our random variable is $D_X := \{H, T\}$. To begin with, we shall define the *probability* of an event to be the *fraction* of times that event occurs out of the total number of trials, in the *limit* that the total number of trials goes to infinity. For example, If we toss the coin many times and 20% of times it lands tails, we say that the probability of tails is 20%. The *probability distribution* for random variable X is a function which maps each value of that random variable to its probability. For our coin example, we can denote the probability distribution by $p(\cdot)$, hence $p(X = H) = .2$ and $p(X = T) = .8$. Note that, based on the definition of probability, we have

$$\sum_{a \in D_X} p(X = a) = 1 \text{ and } \forall a \in D_X : p(X = a) \geq 0$$

When there is no confusion about the domain of the random variable, we overloaded the notation and write $\sum_x p(x) = 1$ where the lowercase letter denotes the values.

We can extend the definition of probability from one variable to *multiple* variables, which results in the notion of *joint* probability distribution. For example, given two random variables X and Y , the joint probability distribution $p(X, Y)$ gives the probability of possible combination of values for these two random variables, hence $\sum_x \sum_y p(x, y) = 1$. We can now define the *marginal* and *conditional* probability distributions, and extend the definitions from *discrete* valued random variables to continuous-valued random variables via *probability density function* where $\int_{-\infty}^{\infty} p(x) dx = 1$. [Appendix A](#) contains the basic concepts of probability theory; please read through it if you feel that you need a refreshment. We now turn our attention to some important concepts from probability theory.

The Sum and product Rules. Given two random variables X and Y , based on the definitions we have had so far, we can write the two fundamental rules of probability theory:

$$\text{Sum Rule: } p(x) = \sum_y p(x, y)$$

Product Rule: $p(x, y) = p(y|x)p(x)$

Bayes Theorem. From the product rule, we immediately obtain the following relationship between conditional probabilities which is called the Bayes' theorem:

$$p(y|x) = \frac{p(x|y)p(y)}{p(x)}$$

Using the sum rule, the denominator in Bayes' theorem can be expressed in terms of the quantities appearing in the numerator:

$$p(x) = \sum_y p(x|y)p(y)$$

We can view the denominator in Bayes' theorem as being the normalisation constant required to ensure that the sum of the conditional probability on the left-hand side of the above equation over all values of Y equals one.

Independence. Two random variables are called independent if and only if

$$p(x, y) = p(x)p(y)$$

For example, if we toss a coin independently two times, the outcome of the first toss X and the outcome of the second toss Y are independent, hence, their joint probability will be the product of their marginal probabilities.

The Coin Example

Suppose we are given a dataset which contains the outcomes of 10 *independent* tosses of the same coin $\mathcal{D} := \{H, T, T, T, T, H, T, T, H, T\}$, where H and T stand for Head and Tail, respectively. We are asked to build a model for this coin tossing and determine the best value for our model parameters.

We can imagine that this is a damaged coin so that the probability of landing heads is not necessarily the same as that of landing tails. We take a simple probabilistic model for our coin, in which the probability of coming head is denoted by the *parameter* w where $0 \leq w \leq 1$, in other words, $p(x = H) = w$. Since the sum of the probabilities over all outcomes must be one, we know that $p(x = T) = 1 - w$.

Fixing our parametric model for the coin, we can now calculate the probability of having the dataset $\mathcal{D}$ generated from the coin. Why may we want to do that? Because we can then choose a value for our coin parameter which maximises the probability of the generated dataset $\mathcal{D}$, which intuitively seems a reasonable thing to do. The probability of the dataset given the model is called the *likelihood*, which plays a central role in machine learning.

Now let's compute the probability of the dataset based on our parametric model for the coin. Since the coin tosses are *independent*, the probability of the dataset is the product of the probability of individual data points:

$$p(\mathcal{D}|\text{coin model}) = \prod_{i=1}^{10} p(x_i|\text{coin model}) = w^3(1-w)^7$$

Note that the likelihood is a function of w , so we need to set its derivative to zero to maximise it. Instead of maximising the likelihood, we can maximise its log denoted by $\mathcal{L}(w)$, which does not change the solution (since log is a monotone function) but is mathematically more convenient:

$$\frac{d\mathcal{L}(w)}{dw} = 0 \Rightarrow \frac{d[3 \log w + 7 \log(1-w)]}{dw} = 0 \Rightarrow \frac{3}{w} - \frac{7}{1-w} = 0 \Rightarrow w = \frac{3}{3+7}$$

You may now think that it was obvious from the beginning that a reasonable value for the coin parameter should have been $3/10$. However, this example is meant to give a rigorous reasoning for this reasonable intuition. Shortly, we will see that *maximum likelihood* is a central parameter estimation technique in machine learning, and we will apply it to more complex probabilistic models in future.

A Probabilistic Approach to Machine Learning

In probabilistic machine learning, we would like to capture our uncertainty about the data using probability theory. As mentioned earlier, the uncertainty arises both through the noise in measurements, as well as through finite size of data sets. We may wish to represent our uncertainty about model prediction, or even the estimated values for model parameters. In the following, we first see a principle, rooted in probability theory, to estimate model parameters. We then see a technique which can be used to formalise our uncertainties about predictions and estimated model parameters.

Maximum Likelihood Principle. The *maximum likelihood* is a widely used parameter estimation technique in which the model parameters is set to the value that maximises the likelihood function $p(\mathcal{D}|\text{model})$. This corresponds to choosing the value of parameters for which the probability of the observed data set is maximised (as seen in the coin example). In the machine learning literature, the negative log of the likelihood function is called an *error function*. Because the negative logarithm is a monotonically decreasing function, maximising the likelihood is equivalent to minimising the error.

Bootstrap for Quantifying Uncertainty. One approach to quantifying the uncertainty is the *bootstrap* technique, which works as follows. Suppose we are given a data set $\mathcal{D}$, and we would like to fit a model with parameters w to this data. We can resort to maximum likelihood, and estimate a value for w which clearly depends on $\mathcal{D}$. Now, we may wonder to some extent our estimated parameters would change if we have an alternate dataset $\mathcal{D}'$. If we do this exercise for several alternate datasets, then we will a *distribution* over our maximum likelihood estimates for the parameters. The higher the variance of this distribution, the more uncertainty we have about our parameter estimates. The bootstrap technique helps us to quantify our uncertainty not only about the parameter estimates but also about the predictions of the model.

Now the question is how to get the alternate datasets. We are given only one initial dataset, and in most cases, we are not allowed to request more datasets. The idea is to generate the alternate datasets from the original one, as follows. Suppose our original data set consists of N data points

$\mathcal{D} = \{x_1, \dots, x_N\}$. We can create a new dataset $\mathcal{D}'$ by drawing N points at *random* from $\mathcal{D}$ with *replacement*. Therefore, some points in $\mathcal{D}$ may be replicated in $\mathcal{D}'$, whereas other points in $\mathcal{D}$ may be absent from $\mathcal{D}'$. This process can be repeated L times to generate L data sets each of size N and each

obtained by sampling from the original data set $\mathcal{D}$. The statistical accuracy of parameter estimates can then be evaluated by looking at the variability of predictions between the different bootstrap data sets.

Bayesian Approach to Machine Learning

So far in this chapter, we have viewed probabilities in terms of the frequencies of random, repeatable events. We shall refer to this as the *classical* or *frequentist* interpretation of probability. Now we turn to the more general *Bayesian* view, in which probabilities provide a quantification of uncertainty.

Consider an uncertain event, for example, whether the moon was once in its own orbit around the sun, or whether the Arctic ice cap will have disappeared by the end of the century. These are not events that can be repeated numerous times in order to define a notion of probability as we did earlier in the context of boxes of fruit. Nevertheless, we will generally have some idea, for example, of how quickly we think the polar ice is melting. If we now obtain fresh evidence, for instance from a new Earth observation satellite gathering novel forms of diagnostic information, we may revise our opinion on the rate of ice loss.

In the field of machine learning, too, it is helpful to have a more general notion of probability. Consider the example of polynomial curve fitting discussed in Section 2. It seems reasonable to apply the frequentist notion of probability to the random values of the observed variables t_n . However, we would like to address and quantify the uncertainty that surrounds the appropriate choice for the model parameters $\mathbf{w}$. We shall see that, from a Bayesian perspective, we can use the machinery of probability theory to describe the uncertainty in model parameters such as $\mathbf{w}$, or indeed in the choice of the model itself.

Prior and Posterior. We capture our assumptions about model parameters $\mathbf{w}$, *before* observing the data, in the form of a *prior* probability distribution $p(\mathbf{w})$. The effect of the observed data $\mathcal{D}$ is expressed through the conditional probability $p(\mathcal{D}|\mathbf{w})$. Bayes' theorem, which takes the form:

$$p(\mathbf{w}|\mathcal{D}) = p(\mathcal{D}|\mathbf{w})p(\mathbf{w})/p(\mathcal{D}) \quad (1.3.1)$$

then allows us to evaluate the uncertainty in $\mathbf{w}$ after we have observed $\mathcal{D}$ in the form of the *posterior* probability $p(\mathbf{w}|\mathcal{D})$. We can state Bayes' theorem in words

$$\text{posterior} \propto \text{likelihood} \times \text{prior}$$

where all of these quantities are viewed as functions of $\mathbf{w}$. The denominator in (1.3.1) is the normalisation constant, which ensures that the posterior distribution on the left-hand side is a valid probability density and integrates to one. Indeed, integrating both sides of (3.1) with respect to $\mathbf{w}$, we can express the denominator in Bayes' theorem in terms of the prior distribution and the likelihood function

$$p(\mathcal{D}) = \int p(\mathcal{D}|\mathbf{w})p(\mathbf{w})d\mathbf{w}$$

Frequentist Vs Bayesian. In both the Bayesian and frequentist paradigms, the likelihood function $p(\mathcal{D}|\mathbf{w})$ plays a central role. However, the manner in which it is used is fundamentally different in the two approaches. In a frequentist setting, $\mathbf{w}$ is considered to be a fixed parameter, whose value is determined by some form of *estimator*, and error bars on this estimate are obtained by considering the

distribution of possible data sets $\mathcal{D}$ (c.f. the bootstrapping technique). By contrast, from the Bayesian viewpoint, there is only a single data set $\mathcal{D}$ (namely the one that is actually observed), and the uncertainty in the parameters is expressed through a probability distribution over $\mathbf{w}$.

Bayesian Machine Learning. We are now ready to see the fundamental rules of Bayesian machine learning:

$$\text{Prediction: } p(x|\mathcal{D}, M) = \int p(x|\mathbf{w}, \mathcal{D}, M) p(\mathbf{w}|\mathcal{D}, M) d\mathbf{w}$$

$$\text{Model Comparison: } p(M|\mathcal{D}) = \frac{p(\mathcal{D}|M)p(M)}{p(\mathcal{D})}$$

where $p(\mathcal{D}|M) = \int p(\mathcal{D}|\mathbf{w}, M) p(\mathbf{w}|M) d\mathbf{w}$.

The Coin Example: A Bayesian View

Now let us revisit our coin example, and assume that we are given a dataset of the outcome of ten coin tosses where all of which turn out to be heads. If we use the maximum likelihood principle, then the parameter of the coin w (i.e. probability of coming heads) turns out to be one! This means that, we would predict the probability of coming tails for a future coin toss is zero, which may seem unreasonable! The fact is that we have a very small dataset consisting of only ten coin tosses and may have been unlucky to get all of them to be heads, so we should not only rely on the data and instead need to capture our *prior belief* as well.

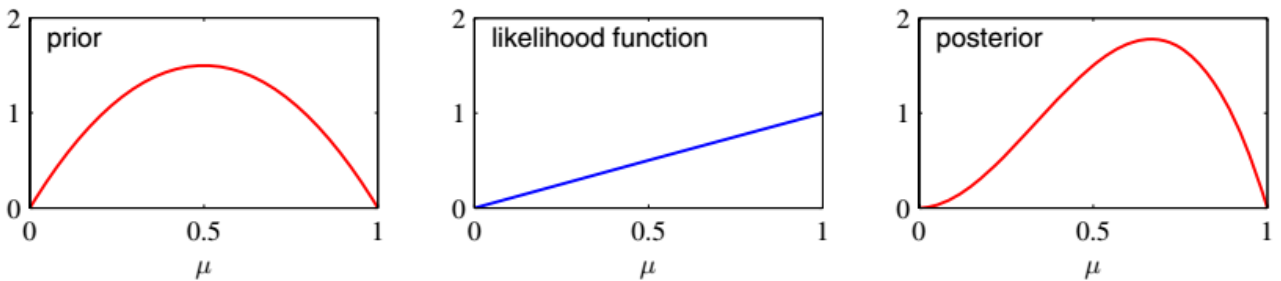


Figure 1.3.1: Illustration of one step of sequential Bayesian inference. The prior is given by a beta distribution with parameters $a = 2, b = 2$, and the likelihood function with $N = m = 1$, corresponds to a single observation of $x = 1$, so that the posterior is given by a beta distribution with parameters $a = 3, b = 2$.

A priori, before seeing any dataset, we may believe that a given coin is most probably fair, i.e. the probability of coming heads and tails are the same and equal .5. Let us assume we use the Beta distribution to represent our prior belief:

$$\text{Beta}(w|a, b) \propto w^{a-1} (1 - w)^{b-1}$$

The Beta distribution has two parameters (a, b) , and puts a distribution over $w \in [0, 1]$ where the mean of the distribution is $\frac{a}{a+b}$. In our case, we may choose to set $(a = 2, b = 2)$ to represent the fact that a priori we believe that most probably a coin is fair, i.e. our prior belief is concentrated around

$\frac{2}{2+2}$. Now we can combine the prior with the likelihood to come up with our updated belief (namely the posterior) about the parameter of the coin (See Figure 1.3.1):

$$\begin{aligned}\text{Posterior} &= \text{Beta}(a, b) \times \text{Likelihood} \\ &\propto w^{a-1} (1-w)^{b-1} \times w^{10} \\ &= w^{a+10-1} (1-w)^{b-1} \\ &= \text{Beta}(a+10, b)\end{aligned}$$

As seen in the last step, the posterior turns out to be a Beta distribution with parameters $(a+10, b)$. So if we set $\text{Beta}(2, 2)$ as our prior, the posterior would be $\text{Beta}(12, 2)$. This means in our updated belief, we mostly believe that the parameter of the coin is $\frac{12}{12+2}$. So in our Bayesian viewpoint we give every value a chance to be the parameter of the coin, where 12/14 is the most probable value.

Interestingly, if we get a new batch of data containing the outcomes of new tosses of the coin, we can start with our current belief $\text{Beta}(10+a, b)$ as the prior, and convert it to the posterior using the likelihood of the new batch of data. This is called the *Bayesian updating*.

Appendix A: Probability Background

Basics of Probability

We will introduce the basic concepts of probability theory by considering a simple example. Imagine we have two boxes, one red and one blue, and in the red box we have 2 apples and 6 oranges, and in the blue box, we have 3 apples and 1 orange. This is illustrated in Figure A.1. Now suppose we *randomly* pick one of the boxes and from that box we *randomly* select an item of fruit, and having observed which sort of fruit it is we replace it in the box from which it came. We could imagine repeating this process many times. Let us suppose that in so doing we pick the red box 40% of the time and we pick the blue box 60% of the time, and that when we remove an item of fruit from a box we are equally likely to select any of the pieces of fruit in the box.

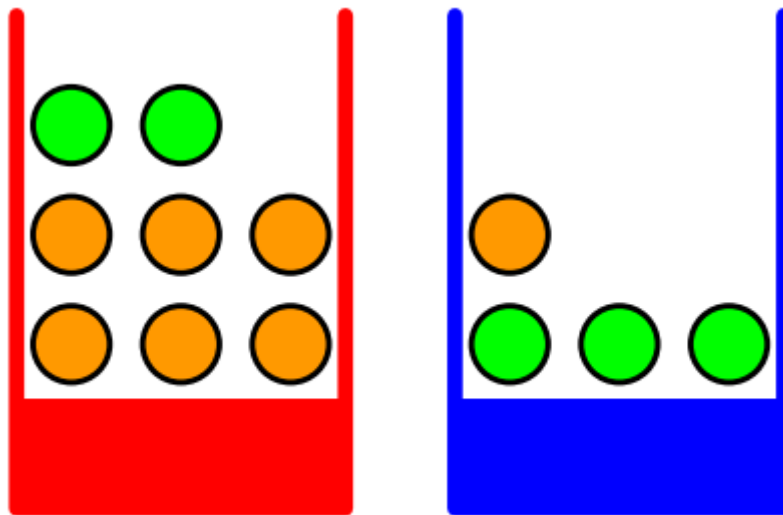


Figure A.1: We use a simple example of two coloured boxes each containing fruit (apples are shown in green and oranges are shown in orange) to introduce the basic ideas of probability.

Random Variable. In this example, the identity of the box that will be chosen is a *random variable*, which we shall denote by B . This random variable can take one of the two possible *values*, namely r (corresponding to the red box) or b (corresponding to the blue box). Similarly, the identity of the fruit is also a random variable and will be denoted by F . It can take either of the values a (for apple) or o (for orange).

Probability. We define the *probability* of an *event* to be the *fraction* of times that event occurs out of the total number of trials, in the limit that the total number of trials goes to infinity. Thus, the probability of selecting the red box is $4/10$, and the probability of selecting the blue box is $6/10$. We write these probabilities as $p(B = r) = 4/10$ and $p(B = b) = 6/10$. Note that, by definition, probabilities must lie in the interval $[0, 1]$. Also, if the events are mutually exclusive and if they include all possible

outcomes (for instance, in this example the box must be either red or blue), then we see that the probabilities for those events must sum to one.

Joint/Marginal/Conditional Probability. Consider the slightly more general example shown in Figure A.2 involving two random variables X and Y (which could for instance be the Box and Fruit variables considered above). We shall suppose that X can take any of the values x_i where $i = 1, \dots, M$, and Y can take the values y_j where $j = 1, \dots, L$. Consider a total of N trials in which we sample both of the variables X and Y , and let the number of such trials in which $X = x_i$ and $Y = y_j$ be denoted by n_{ij} . The probability that X will take the value x_i and Y will take the value y_j is written $p(X = x_i, Y = y_j)$ and is called the *joint probability* of $X = x_i$ and $Y = y_j$. It is defined by the number of points falling in the cell i, j as a fraction of the total number of points:

$$p(X = x_i | Y = y_j) := \frac{n_{ij}}{N}$$

Here we are implicitly considering the limit $N \rightarrow \infty$. Similarly, the probability that X takes the value x_i irrespective of the value of Y , called the *marginal probability* $p(X = x_i)$, is defined by the fraction of the total number of points that fall in column i :

$$p(X = x_i) := \frac{c_i}{N}$$

where c_i is the number of trials in which X takes the value x_i (irrespective of the value that Y takes).

If we consider only those instances for which $X = x_i$, then the fraction of such instances for which $Y = y_j$ is written $p(Y = y_j | X = x_i)$ and is called the *conditional probability*. It is defined by finding the fraction of those points in column i that fall in cell i, j :

$$p(X = x_i, Y = y_j) := \frac{n_{ij}}{c_i}$$

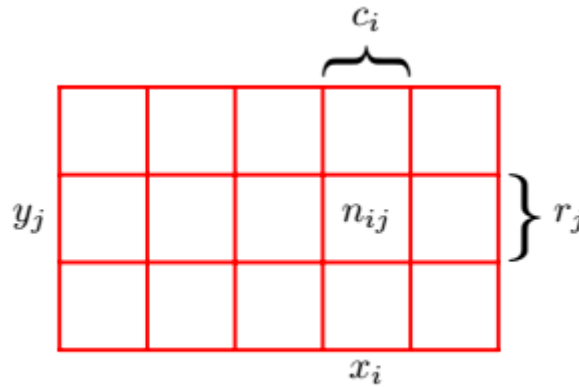


Figure A.2: We can derive the sum and product rules of probability by considering two random variables, X , which takes the values $\{x_i\}$ where $i = 1, \dots, M$, and Y , which takes the values $\{y_j\}$ where $j = 1, \dots, L$. In this

illustration we have $M = 5$ and $L = 3$. If we consider a total number N of instances of these variables, then we denote the number of instances where $X = x_i$ and $Y = y_j$ by n_{ij} , which is the number of points in the corresponding cell of the array. The number of points in column i , corresponding to $X = x_i$, is denoted by c_i , and the number of points in row j , corresponding to $Y = y_j$, is denoted by r_j .

Expectation, Variance, Covariance. One of the most important operations involving probabilities is that of finding weighted averages of functions. The average value of some function $f(x)$ under a probability distribution $p(x)$ is called the *expectation*: $\mathbb{E}[f] := \sum_x p(x)f(x)$.

The variance of $f(x)$ is defined by $\text{var}[f] := \mathbb{E}[(f(x) - \mathbb{E}[f])^2]$. It can be shown that the variance can be re-written as $\mathbb{E}[f(x)^2] - \mathbb{E}[f]^2$.

For two random variables x and y , the covariance is defined by $\text{cov}[x, y] := \mathbb{E}_{x,y}[(x - \mathbb{E}[x])(y - \mathbb{E}[y])]$.

Continuous Random Variables. So far we have seen random variables with discrete values, and the definition of *probability distribution* for them. However, there is another type of random variables which accepts continuous values. We define the *probability density function* for continuous random variables which must satisfy the following conditions

$$p(x) \geq 0 \quad \text{and} \quad \int_{-\infty}^{\infty} p(x)dx = 1$$

The probability that x lies in the interval $(-\infty, z)$ is given by the *cumulative distribution function* defined by

$$P(z) := \int_{-\infty}^z p(x)dx$$

If we have several continuous variables $x_1, \dots, x_D$, denoted collectively by the vector $\mathbf{x}$, then we can define a *joint* probability density $p(\mathbf{x}) = p(x_1, \dots, x_D)$. Accordingly, we can define the conditional and marginal probability density functions, Bayes' theorem, expectation etc for this case.

An example of a famous probability density function for continuous random variables is the *normal* or *Gaussian* distribution. For the case of a single real-valued variable x , the Gaussian distribution is defined by

$$\mathcal{N}(x|\mu, \sigma) := \frac{1}{(2\pi\sigma^2)^{\frac{1}{2}}} \exp\left\{-\frac{1}{2\sigma^2}(x - \mu)^2\right\}$$

where μ and σ are the parameters of the distribution. It can be shown that μ is the expectation and σ^2 is the variance of the Gaussian distribution.

